

Deep Autoencoder-Based Anomaly Detection in Simulated Physics Detector Data

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per la Fisica, 2024–2025

Abstract

We present a deep convolutional autoencoder (CAE) trained exclusively on normal background data to detect anomalous events in simulated 100×100 grayscale detector images. The reconstruction error (mean squared error, MSE) of the trained autoencoder is used as an anomaly score, and detection thresholds are calibrated on a held-out normal validation set for several target false-positive rates (FPR). At a 10% FPR, the model detects 59.53% of events in a high-anomaly-fraction test set and 11.43% of events in a low-anomaly-fraction test set, consistent with the expected anomaly proportions of $> 55\%$ and $< 45\%$, respectively. We further explore unsupervised structure in the 64-dimensional latent space using PCA, UMAP, Gaussian Mixture Models (GMM), and K -Means clustering. While dimensionality-reduction visualizations qualitatively separate high-anomaly samples from normal ones, density-based clustering methods achieve only limited purity, suggesting that reconstruction-error thresholding remains the most reliable anomaly indicator for this dataset.

1 Introduction

Anomaly detection in physics detector data is a central problem when searching for rare or unexpected signals against a large background of well-understood “normal” events. Because anomalous events are, by definition, rare and often unlabeled, supervised classification is not directly applicable. Autoencoders offer a natural unsupervised alternative: a neural network is trained to compress and reconstruct only *normal* data, so that when it is presented with an anomalous sample it is expected to fail to reconstruct it accurately. The resulting reconstruction error can then be used as an anomaly score.

In this work we train a deep convolutional autoencoder on 100×100 grayscale images representing normal background signals from a simulated detector, and evaluate it on two independent test sets with different anomaly fractions. We complement the reconstruction-error analysis with an exploration of the learned latent space using both linear (PCA) and non-linear (UMAP) dimensionality reduction, as well as two clustering algorithms (GMM and K -Means), in order to assess whether unsupervised clustering can recover the anomaly structure directly from the latent representation.

2 Datasets

Three datasets of 100×100 single-channel (grayscale) images were provided:

- **Normal_data:** 12,000 samples of background (normal) detector signals.
- **Test_data_low:** 3,000 samples containing a low mixture of anomalous signals ($< 45\%$ anomalies).

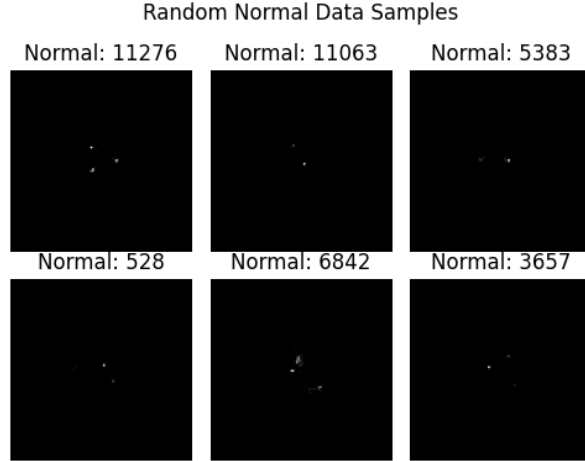


Figure 1: Representative 100×100 samples from the normal background dataset (top) and the high-anomaly-fraction test dataset (bottom).

- **Test_data_high:** 3,000 samples containing a high mixture of anomalous signals ($> 55\%$ anomalies).

The normal dataset was split into a training set (80%, 9,600 samples) and a validation set (20%, 2,400 samples). Only the training split was used to fit the autoencoder; the validation split was reserved to (i) monitor reconstruction quality on unseen normal data and (ii) calibrate the anomaly-score threshold at a chosen false-positive rate.

Figure 1 shows representative samples from the normal and high-anomaly datasets. Visually, the images correspond to sparse, localized measurements rather than complex textured patterns (unlike, e.g., natural images or handwritten digits), and no anomaly pattern is readily apparent to the naked eye, motivating the use of a learned model rather than hand-crafted heuristics.

A design decision was made *not* to normalize the pixel values. While normalization can stabilize PCA and enable a wider choice of loss functions (e.g. binary cross-entropy with a sigmoid output), the pixel intensities here plausibly encode a physical quantity (e.g. deposited energy), and rescaling risks discarding physically meaningful magnitude information. This choice restricted the reconstruction loss to MSE, since a BCE loss combined with a sigmoid output layer would be inconsistent with unnormalized, unbounded input values.

2.1 Preliminary structure via PCA

Before designing the autoencoder, Principal Component Analysis (PCA, 200 components) was applied to the flattened images to obtain an intuition of the global data structure and to estimate a reasonable latent-space dimensionality (Figure 2). The projection onto the first two principal components shows that anomalous samples are shifted towards higher values along the leading component, with the high-anomaly dataset forming a further and more distinctly separated cluster than the low-anomaly dataset. This is consistent with the high-anomaly set containing both a larger number and a more pronounced form of anomalies than the low-anomaly set.

An attempt was made to estimate an appropriate latent-space dimensionality from the PCA explained-variance curve (the “elbow method”). No clear elbow was observed: while the cumulative explained variance does eventually saturate, it does so only around the 175th component, a value considered too large relative to the apparent simplicity of the data. This suggests that PCA, being a linear technique, is not well suited to capturing the (presumably non-linear) structure of this dataset, and an alternative criterion was used to fix the autoencoder’s latent dimension (Section 3).

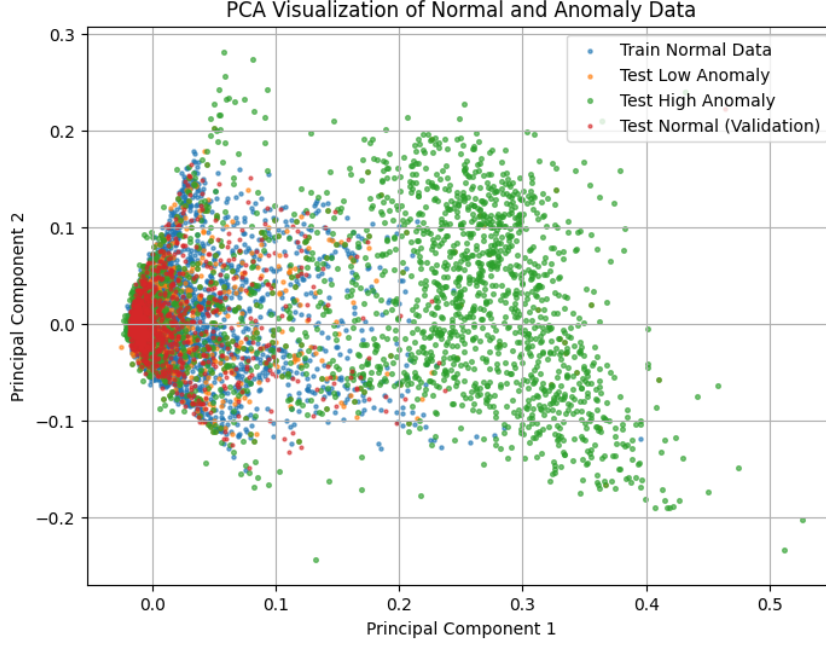


Figure 2: PCA projection (first two components) of the raw pixel data for the training-normal, validation-normal, low-anomaly, and high-anomaly datasets.

3 Model Architecture

A convolutional autoencoder was implemented in PyTorch, consisting of an encoder that compresses each $1 \times 100 \times 100$ image into a 64-dimensional latent vector, and a decoder that reconstructs the image from this latent representation.

3.1 Encoder

The encoder consists of three convolutional layers with 3×3 kernels and stride 2, followed by a fully connected layer projecting onto the latent space:

- Conv2d: $1 \rightarrow 32$ channels, kernel 3×3 , stride 2, ReLU
- Conv2d: $32 \rightarrow 64$ channels, kernel 3×3 , stride 2, ReLU
- Conv2d: $64 \rightarrow 96$ channels, kernel 3×3 , stride 2, ReLU
- Fully connected: flattened features ($96 \times 11 \times 11 = 11,616$) $\rightarrow$ 64-dimensional latent space

Each convolutional layer both downsamples the spatial resolution and increases the number of feature channels, so that spatial information lost through downsampling is compensated for by a richer channel representation. Square kernels and strides were used throughout, since no asymmetry between the horizontal and vertical directions was observed in the input images.

3.2 Decoder

The decoder mirrors the encoder using transposed convolutions:

- Fully connected: $64 \rightarrow 11,616$ dimensions, reshaped to $96 \times 11 \times 11$
- ConvTranspose2d: $96 \rightarrow 64$ channels, kernel 3×3 , stride 2, output padding (1, 1), ReLU
- ConvTranspose2d: $64 \rightarrow 32$ channels, kernel 3×3 , stride 2, ReLU

- ConvTranspose2d: $32 \rightarrow 1$ channel, kernel 3×3 , stride 2, output padding (1, 1)

Output padding of (1,1) was required on selected layers to recover exactly the original 100×100 resolution. The full model, encoder plus decoder, contains approximately 1.65 million trainable parameters (see Table 1).

Table 1: Summary of the autoencoder architecture (input size $1 \times 100 \times 100$).

Layer	Output shape	Parameters
Conv2d (encoder)	$32 \times 49 \times 49$	320
Conv2d (encoder)	$64 \times 24 \times 24$	18,496
Conv2d (encoder)	$96 \times 11 \times 11$	55,392
Linear (latent, $z \in \mathbb{R}^{64}$)	64	743,488
Linear (decoder input)	11,616	755,040
ConvTranspose2d (decoder)	$64 \times 24 \times 24$	55,360
ConvTranspose2d (decoder)	$32 \times 49 \times 49$	18,464
ConvTranspose2d (decoder)	$1 \times 100 \times 100$	289
Total		1,646,849

The latent dimensionality of 64 was chosen as an intermediate value between the raw pixel dimensionality and the (inconclusive) elbow estimate from the PCA analysis, since no clear optimal value could be identified from the explained-variance curve.

4 Training

The autoencoder was trained on the 9,600 normal training images using the Adam optimizer (learning rate 10^{-5}), a batch size of 128, and the MSE reconstruction loss, for 40 epochs. Training was performed on a single NVIDIA Tesla T4 GPU. The relatively small batch size was chosen after preliminary experiments to avoid GPU memory overload and to reduce the risk of underfitting.

The training loss decreased monotonically from 0.0191 at epoch 1 to 0.0009 at epoch 40 (Figure 3, Table 2), with most of the reduction occurring within the first 10 epochs and a slow, steady improvement thereafter. The number of epochs could plausibly have been reduced without a large loss in performance, but 40 epochs were used to trade additional computation time for a more thoroughly converged model.

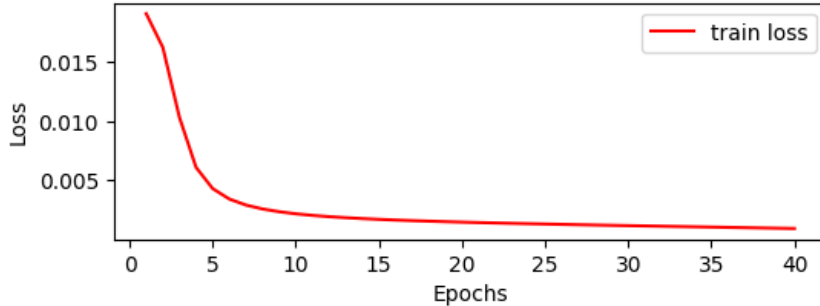


Figure 3: Training loss (MSE) as a function of epoch.

Table 2: Training loss at selected epochs.

Epoch	Loss	Epoch	Loss
1	0.0191	20	0.0015
5	0.0043	25	0.0013
10	0.0022	30	0.0012
15	0.0017	40	0.0009

5 Latent Space Visualization

After training, latent representations $z \in \mathbb{R}^{64}$ were extracted for the training-normal, validation-normal, low-anomaly, and high-anomaly datasets by passing each sample through the encoder only. Two dimensionality-reduction techniques were used to visualize this space in two dimensions.

5.1 PCA of the latent space

A 2-component PCA was fit on the latent representation of the training data and used to project all four datasets (Figure 4). The first two principal components already capture the overwhelming majority of the latent-space variance (99.40% and 0.05%, respectively, for the first and second components), indicating that the learned latent representation is itself highly compressible and effectively low-dimensional along its dominant axis.

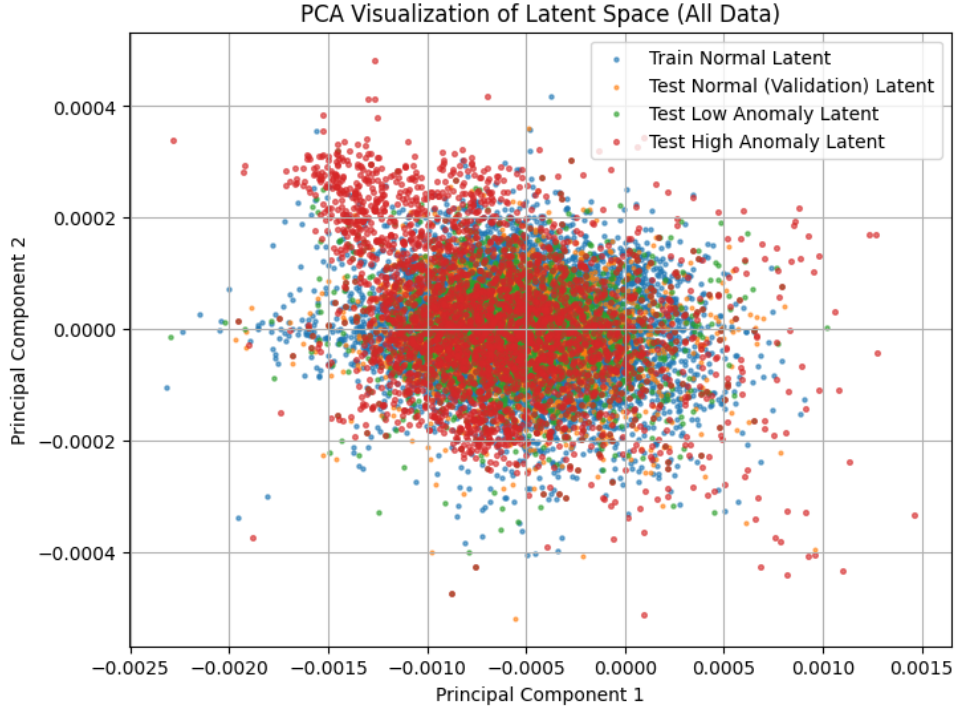


Figure 4: PCA projection of the 64-dimensional latent space for the training-normal, validation-normal, low-anomaly, and high-anomaly datasets.

5.2 UMAP of the latent space

As a non-linear alternative, UMAP was applied to the same latent representations (Figure 5). The resulting embedding shows a qualitatively similar separation between normal and anomalous

samples as the PCA projection, which was taken as evidence that the simpler and computationally cheaper PCA projection is an adequate proxy for the non-linear latent structure in this case.

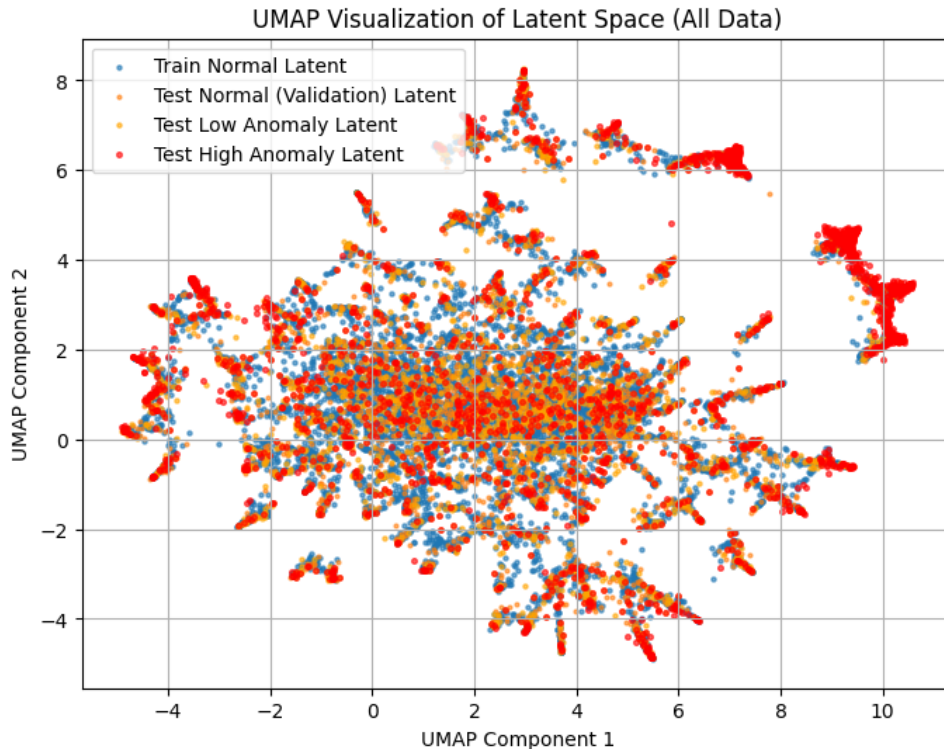


Figure 5: UMAP projection of the 64-dimensional latent space for the training-normal, validation-normal, low-anomaly, and high-anomaly datasets.

6 Anomaly Detection via Reconstruction Error

For each sample x , an anomaly score was defined as the per-sample mean squared reconstruction error,

$$s(x) = \frac{1}{HW} \sum_{i,j} (x_{ij} - \hat{x}_{ij})^2, \quad (1)$$

where $\hat{x} = \text{Decoder}(\text{Encoder}(x))$ and $H = W = 100$. Anomaly scores were computed for the validation-normal set (2,400 samples), the low-anomaly test set (3,000 samples), and the high-anomaly test set (3,000 samples).

An anomaly threshold τ was defined as the percentile of the validation-normal score distribution corresponding to a chosen target false-positive rate (FPR): τ is the $(1 - \text{FPR})$ -th percentile of $\{s(x)\}$ over the normal validation set, so that a fraction FPR of normal samples are (by construction) misclassified as anomalous. Four target FPR values were considered: 10%, 7.5%, 5%, and 2.5%. For each threshold, the fraction of samples exceeding τ was computed separately for the low- and high-anomaly test sets, and is reported as the corresponding “detected anomaly percentage.”

Table 3: Detected anomaly fraction as a function of the target false-positive rate (FPR), calibrated on the normal validation set. Δ is the difference between the high- and low-anomaly detection rates.

Target FPR (%)	Actual FPR (%)	Threshold τ	Anomaly % (High)	Anomaly % (Low)
10.00	10.00	0.000907	59.53	11.43
7.50	7.50	0.000908	56.87	8.53
5.00	5.00	0.000909	54.87	6.37
2.50	2.50	0.000910	50.50	3.33

Figure 6 shows the anomaly-score distributions for the three datasets, together with the four candidate thresholds. The high-anomaly dataset is clearly separated from the normal distribution: a large fraction of its samples occupy a distinct, high-error region, consistent with the expectation from the raw-data PCA (Section 2.1) that its anomalies are both more numerous and more pronounced. The low-anomaly dataset overlaps substantially with the normal distribution, so that only samples with a distinctly higher score than the bulk of normal data are flagged.

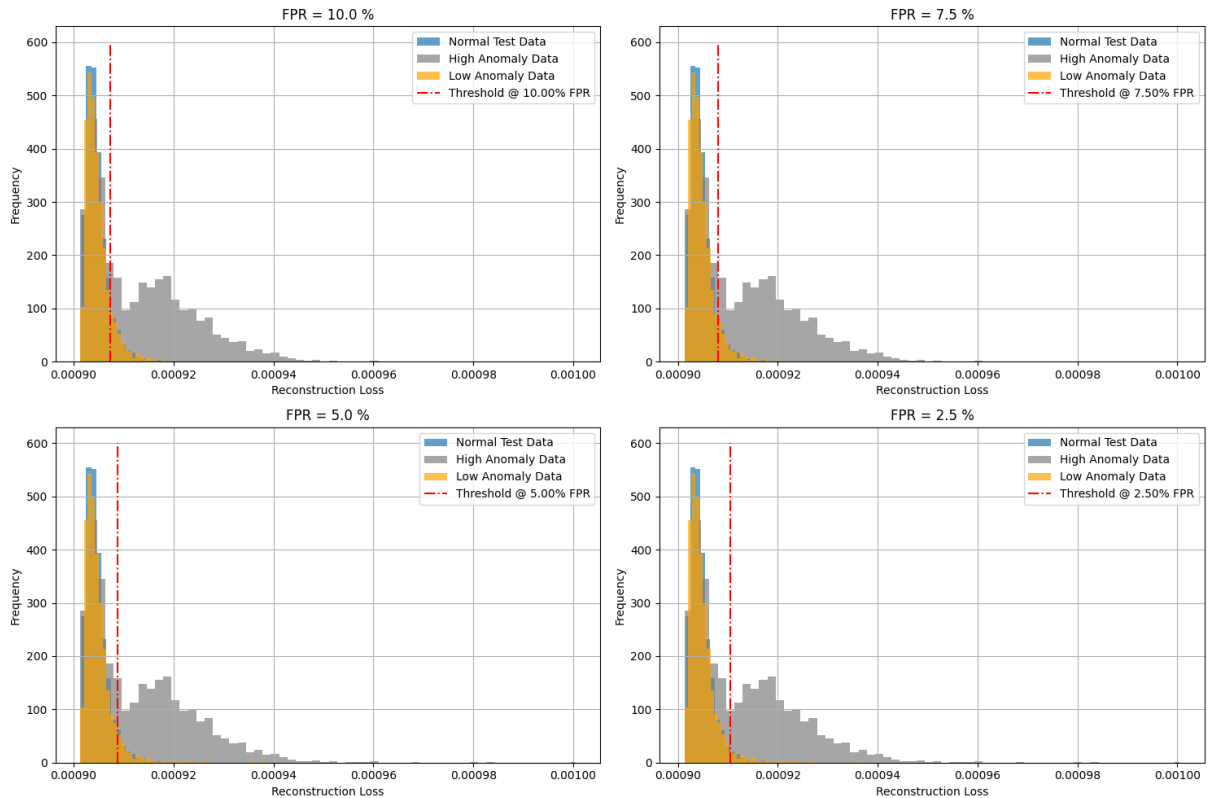


Figure 6: Distribution of the per-sample reconstruction-error anomaly score for the normal validation set, the low-anomaly test set, and the high-anomaly test set, with thresholds corresponding to four target FPR values.

A target FPR of 10% was adopted as the working point, since it gave the best compromise between the two failure modes: using a stricter threshold (e.g. 2.5% or 5%) substantially reduced the number of true anomalies identified (in particular for the low-anomaly set, where the detected fraction dropped to 3.33%), while 10% maximized sensitivity within the maximum allowed FPR. At this working point, the estimated anomaly fractions of 59.53% (high) and 11.43% (low) are consistent with the dataset design specification of $> 55\%$ and $< 45\%$ anomalies, respectively,

supporting the validity of the reconstruction-error approach.

7 Latent Space Clustering

As a complementary, purely unsupervised approach, two clustering algorithms were applied directly to the combined latent representations of all four datasets, without using the reconstruction-error labels.

7.1 Gaussian Mixture Model

A two-component Gaussian Mixture Model (full covariance, up to 10^4 EM iterations) was fit to the combined 64-dimensional latent vectors. The two resulting clusters are only partially aligned with the normal/anomaly distinction. Examining the composition of the cluster enriched in anomalous samples (Table 4), 43.29% of its members are genuine anomalies (low- or high-anomaly test samples), i.e. a purity substantially above the overall anomaly base rate, but only 30.58% of the total anomalous population is captured by this cluster. In other words, GMM correctly enriches for anomalies but at the cost of missing the majority of them.

Table 4: Composition of the anomaly-enriched GMM cluster.

Source dataset	Count	Fraction of cluster
Train Normal	1,911	45.08%
Test Normal (val.)	493	11.63%
Test Low Anomaly	625	14.74%
Test High Anomaly	1,210	28.54%
Total anomalous (low + high)	1,835	43.29%

Multiple hyperparameter configurations were explored, including allowing full covariance matrices for additional flexibility, and the same procedure was repeated on the UMAP embedding, without qualitative improvement. This suggests that the limiting factor is not the choice of clustering algorithm or projection, but rather the degree to which the learned latent space itself linearly (or Gaussian-mixture-wise) separates normal from anomalous samples.

7.2 K -Means

A K -Means clustering with $k = 5$ clusters was also attempted on the same combined latent representations, motivated by the expectation that the data may not be well described by simple circular (isotropic Gaussian) clusters. This approach did not yield clusters that meaningfully separated normal from anomalous samples, and no anomaly-purity analysis analogous to Table 4 could be usefully extracted from it.

8 Discussion

Two complementary strategies were used to identify anomalous samples: (i) a global reconstruction-error threshold, and (ii) unsupervised clustering of the latent representation. The reconstruction-error approach performed considerably better: at the working point of 10% FPR, it recovered detection rates consistent with the known anomaly fractions of both test sets, whereas latent-space clustering (GMM or K -Means) could only identify a minority ($\sim 31\%$) of the anomalous population, albeit with an enriched purity ($\sim 43\%$ anomalous within the anomaly-associated cluster, versus the raw dataset composition).

This discrepancy suggests that while the autoencoder’s latent space does encode information that correlates with anomalous behavior, the correlation is not strong enough to produce well-separated, convex clusters under a Gaussian-mixture or centroid-based model. This is consistent with the raw-data PCA analysis, which likewise failed to find a well-defined “elbow” or clearly separated linear structure. It is possible that the latent space, being optimized purely for reconstruction fidelity rather than for anomaly separability, does not induce a geometry favorable to standard clustering algorithms; architectures or training objectives that explicitly encourage a more separable or disentangled latent space (e.g. contrastive or variational objectives with anomaly-aware regularization) could be a promising direction for future work.

9 Conclusions

A convolutional autoencoder trained solely on normal detector data was shown to be an effective anomaly detector when anomaly scores are derived from the per-sample reconstruction error. Using a calibrated threshold at 10% false-positive rate, the model detected 59.53% of samples in the high-anomaly-fraction dataset (specification: $> 55\%$ anomalies) and 11.43% of samples in the low-anomaly-fraction dataset (specification: $< 45\%$ anomalies), in both cases consistent with the expected anomaly composition of the datasets and supporting the model’s ability to capture meaningful structural differences between normal and anomalous events.

By contrast, clustering directly in the 64-dimensional latent space (via GMM or K -Means, with or without a prior PCA/UMAP projection) achieved only limited success, correctly capturing about 31% of the total anomalous population within a cluster whose anomaly purity was $\sim 43\%$. This indicates that, while informative, the learned latent space does not by itself yield a natural separation into normal/anomalous groups under standard unsupervised clustering assumptions.

Overall, the reconstruction-error-based approach constitutes a solid baseline for this anomaly detection task, and the results provide a good starting point for future work aimed at improving both the model architecture (e.g. deeper or attention-based encoders, or an explicit anomaly-aware regularization of the latent space) and the training strategy, in order to increase the accuracy and robustness of the system.

Acknowledgments

This project was carried out for the *Metodi di Intelligenza Artificiale e Machine Learning per la Fisica* course, Sapienza Università di Roma, 2024–2025, following the guidelines provided at: https://github.com/stefanogiagu/corso_AI_2025/blob/main/ProgettoGruppo/MetodiAI_Fis2025_GroupProject.ipynb